

SYED MUHAMMAD RAFAY ALI

Full Stack Software Engineer

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PROFESSIONAL SUMMARY

Full-stack software engineer and Computer Science graduate from FAST-NUCES with hands-on experience shipping full-stack web applications using React, TypeScript, Node.js, and Express. Proven ability to design and deploy RESTful APIs, integrate third-party services, and build CI/CD pipelines with GitHub Actions and Docker across five end-to-end projects spanning booking platforms, AI-powered event management, and interactive 3D interfaces. Additional exposure to machine learning techniques, including neural networks and K-Means clustering, applied to real-world route optimization problems.

EDUCATION

National University of Computer and Emerging Sciences (FAST-NUCES)

2022 – 2026

Bachelor of Science in Computer Science

TECHNICAL SKILLS

Languages: C/C++, JavaScript, TypeScript, Python

Frontend: React, React Router, TanStack Query, HTML5, CSS3, Tailwind CSS, Bootstrap, jQuery, Vite, Responsive Design

Backend: Node.js, Express.js, RESTful APIs, JWT, API Development, Data Modeling

Databases & DevOps: MongoDB, PostgreSQL, Supabase, Docker, GitHub Actions, CI/CD, Git, Vercel, Railway, Jest, Supertest

CERTIFICATIONS

[Meta Full Stack Developer: Front-End & Back-End from Scratch](#) | Meta / Coursera

[Prompt Engineering](#) | Coursera

PROJECTS

HostelSpot: Student Hostel Booking Platform | *React, TypeScript, Node.js, Supabase (PostgreSQL), Tailwind CSS, Docker, GitHub Actions*

- Built a full-stack hostel booking platform with dual dashboards for students and owners, real-time availability updates, and an admin panel; designed a responsive, cross-browser UI.
- Developed RESTful APIs integrated with Supabase (PostgreSQL) and crafted data models for authentication and real-time data sync, reducing booking confirmation time by ~60%.
- Configured a CI/CD pipeline using GitHub Actions and Docker to automate testing and deployment.

FestivAI: AI-Powered Event Management Platform | *React, TypeScript, Node.js, Express, MongoDB, Google Maps API, JWT*

- Engineered a full-stack event-management platform featuring RSVP-driven booking, dynamic QR check-ins, venue management, and AI-generated e-card invitations.
- Integrated Google Maps API for real-time hall location discovery and built RESTful API endpoints connecting the React frontend to a MongoDB backend.
- Implemented JWT-based authentication and role-based access control to separate organizer and attendee permissions.

CarInsight Pro: Car Research & Comparison Platform | *React, Node.js, Express, MongoDB, GitHub Actions, Jest, Supertest, Railway*

- Built a MERN-stack platform for side-by-side car specification comparisons with an interactive matchmaking quiz and a custom CMS.
- Designed RESTful API routes serving structured car data with filtering, sorting, and comparison queries; ran 7 Jest/Supertest integration tests against an in-memory MongoDB instance on every push before deployment to Railway.

PakTravel AI: Intercity Bus Travel System | *Python, Pandas, NumPy, scikit-learn, Neural Networks, K-Means Clustering*

- Built an AI-powered intercity bus travel system combining classical search algorithms, propositional logic, and CSP-based scheduling to model real-world route constraints.
- Applied neural networks and K-Means clustering for route optimization and demand analysis, processing structured travel datasets with Pandas and NumPy to train and validate models across multiple route configurations.

Interactive 3D Portfolio | *TypeScript, SVG, Custom Shaders, CSS Animations*

- Built a visually rich personal portfolio featuring a custom mathematical SVG character with cursor-tracking eyes, custom shaders, and scroll-based reveal animations, all without third-party animation libraries.
- Engineered smooth cross-device performance by optimizing render cycles and implementing lazy-loaded sections.